Experiment 2 — VHDL Code for 4-bit Universal Shift Register

Tools: Xilinx Vivado 2018.2 • Artix-7 Basys 3 (xc7a35tcpg236-1)

1. VHDL Design Code (shiftreg.vhd)

```
----------------------------------------
-- Module Name : shiftreg - Behavioral
-- Design      : 4-bit Universal Shift Register
----------------------------------------
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity shiftreg is
    Port ( clk  : in  STD_LOGIC;
           rst  : in  STD_LOGIC;
           load : in  STD_LOGIC;                    -- parallel load enable
           si   : in  STD_LOGIC;                    -- serial input
           pi   : in  STD_LOGIC_VECTOR (3 downto 0); -- parallel input
           mode : in  STD_LOGIC_VECTOR (1 downto 0); -- mode select
           po   : out STD_LOGIC_VECTOR (3 downto 0); -- parallel output
           so   : out STD_LOGIC);                   -- serial output
end shiftreg;

architecture Behavioral of shiftreg is
    signal reg : STD_LOGIC_VECTOR (3 downto 0);
begin
    process(clk, rst)
    begin
        if rst = '1' then
            reg <= "0000";
        elsif rising_edge(clk) then
            if load = '1' then
                reg <= pi;                  -- synchronous parallel load
            else
                case mode is
                    when "00" =>            -- hold (no operation)
                        reg <= reg;
                    when "01" =>            -- shift right : si -> MSB
                        reg <= si & reg(3 downto 1);
                    when "10" =>            -- shift left  : si -> LSB
                        reg <= reg(2 downto 0) & si;
                    when others =>          -- "11" : parallel load
                        reg <= pi;
                end case;
            end if;
        end if;
    end process;

    po <= reg;
    so <= reg(0);                           -- serial output (LSB)
end Behavioral;
```

2. Constraints File (shiftreg.xdc / UCF)

```
## Constraints file (shiftreg.xdc) – Basys 3 / Artix-7 (xc7a35tcpg236-1)

## Clock (100 MHz on-board oscillator)
set_property PACKAGE_PIN W5  [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports clk]
create_clock -period 10.00 -name sys_clk [get_ports clk]

## Reset (centre push button, btnC)
set_property PACKAGE_PIN U18 [get_ports rst]
set_property IOSTANDARD LVCMOS33 [get_ports rst]

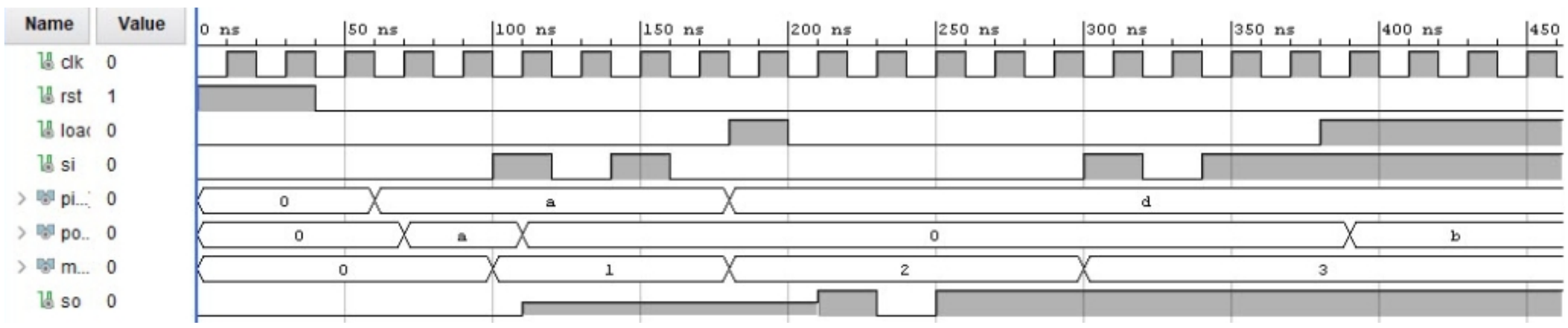
## Control / data inputs (slide switches)
set_property PACKAGE_PIN V17 [get_ports load]
```

```
set_property PACKAGE_PIN V16 [get_ports si]
set_property PACKAGE_PIN W16 [get_ports {pi[0]}]
set_property PACKAGE_PIN W17 [get_ports {pi[1]}]
set_property PACKAGE_PIN W15 [get_ports {pi[2]}]
set_property PACKAGE_PIN V15 [get_ports {pi[3]}]
set_property PACKAGE_PIN W14 [get_ports {mode[0]}]
set_property PACKAGE_PIN W13 [get_ports {mode[1]}]

##Outputs  (LEDs)
set_property PACKAGE_PIN U16 [get_ports {po[0]}]
set_property PACKAGE_PIN E19 [get_ports {po[1]}]
set_property PACKAGE_PIN U19 [get_ports {po[2]}]
set_property PACKAGE_PIN V19 [get_ports {po[3]}]
set_property PACKAGE_PIN W18 [get_ports so]

##I/O standard for all remaining ports
set_property IOSTANDARD LVCMOS33 [get_ports load]
set_property IOSTANDARD LVCMOS33 [get_ports si]
set_property IOSTANDARD LVCMOS33 [get_ports {pi[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {pi[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {pi[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {pi[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {mode[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {mode[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {po[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {po[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {po[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {po[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports so]
```

3. Simulation Waveform



4. VHDL Testbench Code (shifreg_tb.vhd)

```
------------------------------------------------
-- Testbench : shifreg_tb.vhd  (for shiftreg)
------------------------------------------------
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity shifreg_tb is
end shifreg_tb;

architecture Behavioral of shifreg_tb is

    component shiftreg
        Port ( clk  : in  STD_LOGIC;
               rst  : in  STD_LOGIC;
               load : in  STD_LOGIC;
               si   : in  STD_LOGIC;
               pi   : in  STD_LOGIC_VECTOR (3 downto 0);
               mode : in  STD_LOGIC_VECTOR (1 downto 0);
               po   : out STD_LOGIC_VECTOR (3 downto 0);
               so   : out STD_LOGIC);
    end component;

    signal clk,rst,load,si,so : STD_LOGIC := '0';
    signal pi,po : STD_LOGIC_VECTOR (3 downto 0) := (others => '0');
    signal mode  : STD_LOGIC_VECTOR (1 downto 0) := "00";
    constant Tclk : time := 10 ns;

begin

    uut : shiftreg port map (
        clk => clk, rst => rst, load => load, si => si,
        pi => pi, mode => mode, po => po, so => so );

    -- clock generation
    clk_proc : process
    begin
        clk <= '0'; wait for Tclk/2;
        clk <= '1'; wait for Tclk/2;
    end process;

    -- stimulus
    stim_proc : process
    begin
        rst <= '1';  wait for 30 ns;  rst <= '0';

        pi <= "1010"; load <= '1'; wait for Tclk;  load <= '0'; -- load a
        mode <= "01"; si <= '1';   wait for 4*Tclk;           -- shift right
        mode <= "10"; si <= '0';   wait for 4*Tclk;           -- shift left
        pi <= "1101"; load <= '1'; wait for Tclk;  load <= '0'; -- load d
        mode <= "00";              wait for 3*Tclk;           -- hold
        mode <= "11";              wait for 4*Tclk;           -- parallel load
        wait;
    end process;

end Behavioral;
```

5. Synthesis Report (Selected Pages)

```
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------------------------------------------------------------------------------------
| Tool Version : Vivado v.2018.2 (win64) Build 2258646
| Date         : Fri Aug  7 12:10:22 2026
| Host         : DESKTOP-22PHA3V running 64-bit
| Command      : report_utilization -file shiftreg_utilization_synth.rpt
| Design       : shiftreg
| Device       : 7a35tcpg236-1
| Design State : Synthesized
------------------------------------------------------------------------------------

Utilization Design Information

1. Slice Logic
--------------
+-------------------------+------+-------+-----------+-------+
|        Site Type        | Used | Fixed | Available | Util% |
+-------------------------+------+-------+-----------+-------+
| Slice LUTs*             |    5 |     0 |     20800 |  0.02 |
|   LUT as Logic          |    5 |     0 |     20800 |  0.02 |
|   LUT as Memory         |    0 |     0 |      9600 |  0.00 |
| Slice Registers         |    4 |     0 |     41600 |  0.01 |
|   Register as Flip Flop |    4 |     0 |     41600 |  0.01 |
|   Register as Latch     |    0 |     0 |     41600 |  0.00 |
| F7 Muxes                |    0 |     0 |     16300 |  0.00 |
| F8 Muxes                |    0 |     0 |      8150 |  0.00 |
+-------------------------+------+-------+-----------+-------+

1.1 Summary of Registers by Type
--------------------------------
+-------+--------------+-------------+--------------+
| Total | Clock Enable | Synchronous | Asynchronous |
+-------+--------------+-------------+--------------+
| 0     |            _ |           - |            - |
| 0     |            _ |           - |          Set |
| 4     |            _ |           - |        Reset |
| 0     |            _ |         Set |            - |
| 0     |          Yes |           - |            - |
+-------+--------------+-------------+--------------+

4. IO and GT Specific
---------------------
+-----------------------------+------+-------+-----------+-------+
|          Site Type          | Used | Fixed | Available | Util% |
+-----------------------------+------+-------+-----------+-------+
| Bonded IOB                  |   15 |     0 |       106 | 14.15 |
| Bonded IPADs                |    0 |     0 |        10 |  0.00 |
| PHY_CONTROL                 |    0 |     0 |         5 |  0.00 |
| IDELAYCTRL                  |    0 |     0 |         5 |  0.00 |
| IBUFDS                      |    0 |     0 |       104 |  0.00 |
| ILOGIC                      |    0 |     0 |       106 |  0.00 |
| OLOGIC                      |    0 |     0 |       106 |  0.00 |
+-----------------------------+------+-------+-----------+-------+

5. Clocking
-----------
+------------+------+-------+-----------+-------+
|  Site Type | Used | Fixed | Available | Util% |
+------------+------+-------+-----------+-------+
| BUFGCTRL   |    1 |     0 |        32 |  3.13 |
| BUFIO      |    0 |     0 |        20 |  0.00 |
| MMCME2_ADV |    0 |     0 |         5 |  0.00 |
| PLLE2_ADV  |    0 |     0 |         5 |  0.00 |
+------------+------+-------+-----------+-------+

7. Primitives
-------------
+----------+------+---------------------+
| Ref Name | Used | Functional Category |
+----------+------+---------------------+
| FDCE     |    4 |       Flop & Latch  |
| IBUF     |   10 |                  IO |
```

```
|OBUF     |    5|              IO|
|LUT6     |    3|             LUT|
|LUT3     |    2|             LUT|
|BUFG     |    1|           Clock|
+---------+-----+----------------+

8. Black Boxes       : none
9. Instantiated Netlists : none
```

6. Implementation — I/O Planning & Device View

